

Michael Dresser

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SKYDIO SR. AUTONOMY ENGINEER → STAFF AUTONOMY ENGINEER
May 2024 - Present | San Mateo, CA

At Skydio I bridged the gaps between customer-facing cloud software, cloud infra, robotics middleware, and Linux systems to design and deliver complete solutions for internal needs and for our customers.

- Led multi-org rollout of Foxglove. Built a streaming log-conversion service, reducing time-to-visualization from 5+ minutes to 20s. Managed a 6+ month internal roadmap and the vendor relationship. Partnered with autonomy teams to onboard them to Foxglove. (C++, Rust, Typescript, AWS, K8s)
- Owned core simulator infrastructure for multi-SOC robotics platforms, enabling scaled deployment to customers (100s/day) and accurate filesystem/network emulation. (Linux, C++, Python, K8s)
- Ran a program for automated real-world and simulation testing of flight software. Executed >700 real-world flights and 80,000 simulator flights per month. Managed a test operator and organized the work of a small team of engineers.

KUBECOST SW ENGINEER → SR. → STAFF SOFTWARE ENGINEER
Jan 2021 - April 2024 | Remote

- 100x'd supportable scale by proposing, POCing, and productionizing a backend architecture overhaul. Enabled sales into a previously-inaccessible market segment. (Go, DuckDB)
- Scaled quantile-based container sizing recommendations to 250,000+ container environments. (Go)
- Built and marketed an open-source CLI. 700+ GitHub stars. (Go)
- Designed a filtering language for core APIs. Built a lexer, parser, and interpreter. (Go)
- Enabled upgrades without data loss by leading design and implementation of schema backwards-compatibility. (Go)
- Reduced developer feedback loop from 10 minutes to 30 seconds by optimizing builds and enabling local runs. (Go, Docker)

GOOGLE SOFTWARE ENGINEERING INTERN
May 2020 - Aug 2020 | Remote

- Contributed to Kubernetes test infra, simplifying E2E test workflow in an environment running >90,000 tests/month. (Go)

TWILIO SOFTWARE ENGINEERING INTERN
May 2019 - Aug 2019 | Denver, CO | Go, Kubernetes

LOGRHYTHM SOFTWARE ENGINEERING INTERN
Jun 2018 - Aug 2018 | Boulder, CO | Go

CONFERENCE PRESENTATIONS

NODE SIZE MATTERS

"Running K8s as Cheaply as Possible"
KubeCon NA 2023 (November)
Michael Dresser and Alex Meijer
[Link](#)

SKILLS

LANGUAGES

Go • Python • Bash • SQL • Rust • C++ • TypeScript • Clojure • JavaScript

TOOLS

Linux • Git • Kubernetes • Docker • DuckDB • Protobuf • SQLite • React • GCP • AWS

VOLUNTEERING

GUNN ROBOTICS TEAM

2023 - Present
Mentor for a high school FIRST Robotics Competition (FRC) team. I teach (CNC) machining, software, and leadership skills.

EDUCATION

CU BOULDER

B.S. & M.S. IN COMPUTER SCIENCE
GPA: 4.0 / 4.0

SELECT PROJECTS

EVE ITEM DIFF

Rust, WASM
Browser tool for parsing and diffing lists of items in EVE Online's variety of text formats.
Hand-rolled recursive-descent parser. [Link](#).

THESIS: ARUGULA

Scala
A probabilistic programming language. It has special semantics for delayed sampling combined with symbolic inference.